Architecture & MCU Selection

Marklin WiFi AC Train Controller

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Test Project: Architecture & MCU Selection

Candidate Comparison

Feature	ESP32-C3	ESP32-C6	ESP32-S3
CPU	RISC-V single-core, 160 MHz	RISC-V single-core, 160 MHz + LP core 20 MHz	Xtensa LX7 dual-core, 240 MHz
RAM	400 KB SRAM	512 KB + 16 KB (LP)	512 KB SRAM
Flash	4 MB (typical)	4-16 MB	External (module-dependent, 4 MB+)
WiFi	802.11 b/g/n	802.11 ax (WiFi 6) + b/g/n	802.11 b/g/n
Bluetooth	BLE 5.0	BLE 5.3	BLE 5.0
Zigbee/Thread	No	Yes (802.15.4)	No
GPIO	22 (11 usable)	22-30 (package dependent)	Up to 45
ADC	2x 12-bit, 6 channels	12-bit, 8 channels	2x 12-bit, 20 channels
PWM	6 LED PWM channels	6 LED PWM channels	8 LED PWM channels
Native USB	Yes (CDC/JTAG)	Yes (CDC)	Yes (CDC/JTAG)
Package (smallest)	QFN32, 5x5 mm	QFN32, 5x5 mm	QFN56, 7x7 mm
Active WiFi	95-240 mA	~60 mA typical	78-240 mA
Deep sleep	~5 uA	~7 uA	~9 uA
Temp range	-40 to +85C (105C ext.)	-40 to +85C (105C ext.)	-40 to +85C (105C ext.)
LCSC chip price	~$1.15	~$2.18	~$2.14
LCSC module price	~$2.17-2.43	~$2.83-3.24	~$3.46-3.80
PlatformIO	Full support	Partial (needs community fork)	Full support

Assessment Against Requirements

Real-time requirements

- Motor PWM control: All three candidates have sufficient PWM channels (need 2 for motor windings)
- Modbus TCP server: Single-core at 160 MHz is sufficient; dual-core is not needed
- No hard real-time constraints identified

Memory requirements

- Modbus TCP + WiFi stack: ~200-300 KB RAM typical
- All three have sufficient RAM (400-512 KB)
- 4 MB flash is adequate for firmware

Power requirements

- All power from 24 VAC track. After rectification: ~30-34 VDC peak
- Need buck converter to 3.3V for MCU
- WiFi active current 60-240 mA at 3.3V = 0.2-0.8W - negligible vs motor power

- ESP32-C6 has lowest active WiFi consumption (~60 mA typical) due to WiFi 6

Package size

- Must fit inside locomotive body. Compact PCB is critical
- ESP32-C3 and C6: 5x5 mm QFN - smallest
- ESP32-S3: 7x7 mm QFN - larger, but still feasible
- Module size matters more than bare chip (antenna integration)

Cost

- ESP32-C3 is cheapest ($1.15 chip, $2.17 module)
- ESP32-S3 and C6 are comparable ($2.14-2.18 chip)
- For a hobby/low-volume project, module price difference is negligible

Toolchain maturity

- ESP32-C3: Mature, full support across ESP-IDF, Arduino, PlatformIO
- ESP32-S3: Mature, full support
- ESP32-C6: ESP-IDF and Arduino supported, but PlatformIO requires community fork - less mature

Recommendation: ESP32-C3

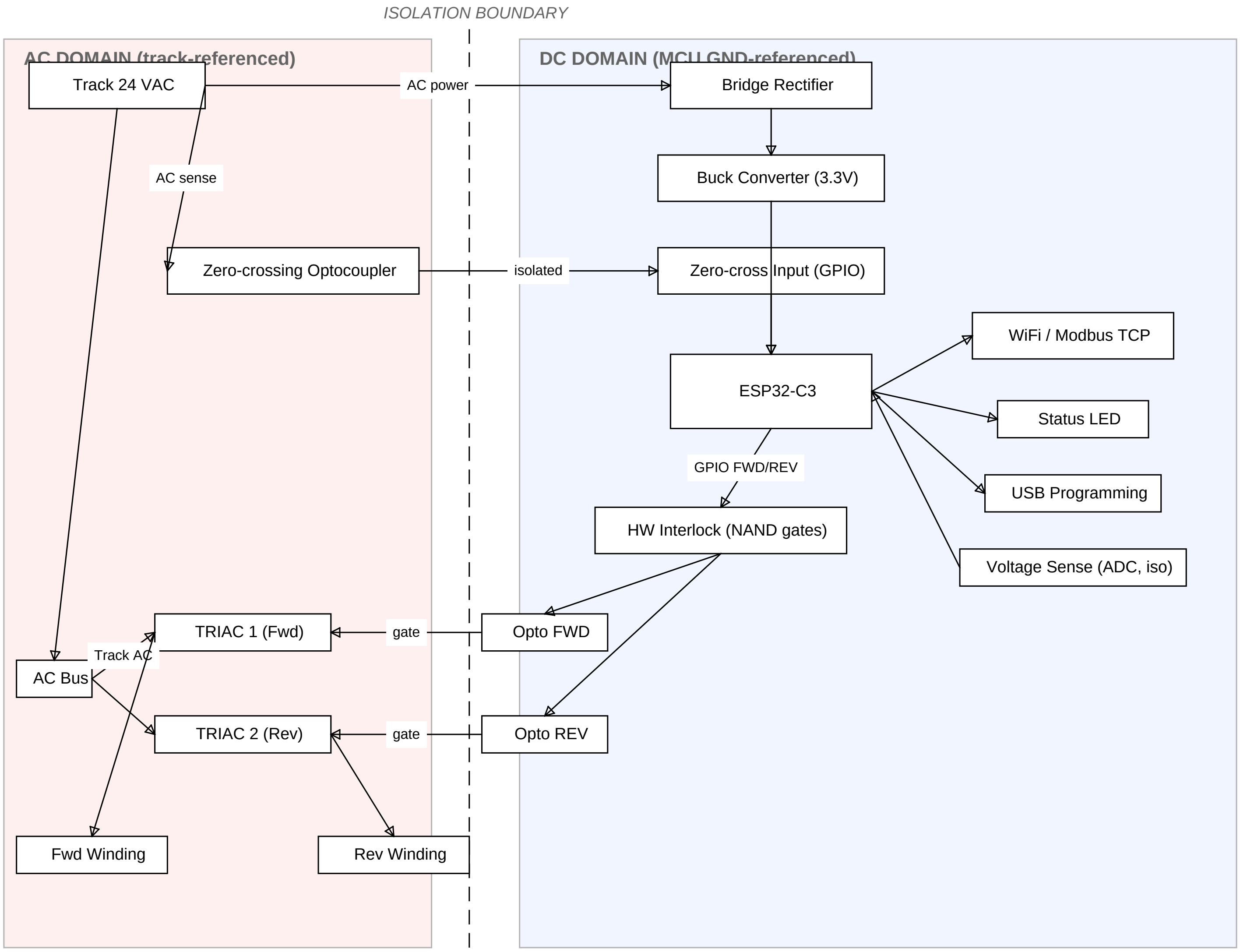
Rationale:

1. Sufficient performance - 160 MHz RISC-V core handles Modbus TCP + PWM motor control with margin. Dual-core (S3) is overkill for this application.
2. Smallest and cheapest - 5x5 mm QFN package, $1.15 chip / $2.17 module. Important for fitting inside a locomotive.
3. Most mature toolchain - Full ESP-IDF, Arduino, and PlatformIO support. No workarounds needed (unlike C6 PlatformIO issues).
4. Native USB - Built-in CDC/JTAG eliminates need for an external USB-UART bridge (e.g., CP2102), simplifying BOM and PCB.
5. Adequate peripherals - GPIO for TRIAC gate triggers, zero-crossing detection input, ADC for voltage sensing, UART for debug, GPIO for LED.
6. WiFi 6 not needed - The C6's WiFi 6 and Thread/Zigbee support offer no benefit for a Modbus TCP application on a home network.

Candidates not selected:

- ESP32-S3: Dual-core 240 MHz is more CPU than needed. Larger package (7x7 mm). Higher module cost. The extra GPIO and ADC channels are not required.
- ESP32-C6: WiFi 6 and Zigbee/Thread are not needed. PlatformIO support is immature (requires community fork). Slightly more expensive than C3 with no practical benefit for this application.

System Architecture Overview



Two separate ground domains:

- AC domain: Track-referenced. TRIACs, motor windings, and zero-crossing sense circuit operate here
- DC domain: MCU GND, derived from bridge rectifier output. ESP32, LED, USB operate here
- Isolation boundary: Optocouplers bridge the two domains for TRIAC gate drive and zero-crossing detection

Motor drive approach:

- The motor is AC-driven - the field windings receive track AC directly, NOT rectified DC
- Direction control: Two TRIACs, one per field winding. Only one is active at a time. The ESP32 selects direction by triggering the appropriate TRIAC gate via optocoupler isolation
- Hardware interlock: A hardware mutual exclusion circuit (e.g., cross-coupled NAND gates or similar logic) prevents both TRIAC gate signals from being active simultaneously, regardless of firmware state. This protects the motor even in case of MCU software fault or runaway
- Speed control: Phase-angle control - the ESP32 detects AC zero-crossings via isolated optocoupler interrupt, then fires the TRIAC gate at a variable delay to control the portion of each AC half-cycle delivered to the motor
- Zero-crossing detection: An optocoupler on the track AC provides galvanic isolation and a clean logic-level signal to an ESP32 GPIO input

Key architectural decisions:

- AC motor path is galvanically separate from the DC MCU domain - optocouplers provide isolation for all signals crossing the boundary
- Single 3.3V rail for MCU from bridge rectifier + buck converter

- Hardware interlock gate logic between ESP32 GPIOs and TRIAC optocouplers - ensures mutual exclusion of forward/reverse windings independent of software
- Three optocouplers needed: 2x TRIAC gate drive, 1x zero-crossing detection
- Zero-crossing detector needed for phase-angle speed control
- TRIACs chosen over relays: no mechanical wear, fast switching, silent operation, enables phase-angle speed control (relays can only do on/off)
- Voltage sensing of track AC also requires isolation (isolated ADC input or optocoupler-based sensing)
- USB native on ESP32-C3 used for programming - no external UART bridge IC needed